

A PROJECT TITLE
ON
Soil Quality Classification Using Random Forest and
Decision Tree

project report submitted in partial fulfilment of the requirements for the Course

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Abstract

This project presents a machine learning system for classifying soil quality using two supervised tree-based algorithms, Decision Tree and Random Forest. The study covers data collection and preprocessing of soil parameters such as pH, nitrogen, phosphorus, potassium, organic matter and moisture, followed by missing-value imputation, encoding and feature engineering. A Decision Tree is first built using Gini impurity and entropy-based splitting, with pruning to control overfitting and manage the bias–variance trade-off. A Random Forest ensemble, using bagging and feature subsampling, is then trained to reduce variance and improve generalization on unseen soil samples. Models are evaluated using accuracy, precision, recall, F1-score and the confusion matrix, with cross-validation used to obtain reliable results and prevent data leakage. Feature-importance analysis identifies the soil attributes that most influence classification, and the comparative study examines whether the ensemble Random Forest improves on a single Decision Tree in accuracy and stability for real-world soil quality prediction.

Keywords: Soil Quality Classification, Random Forest, Decision Tree, Precision Agriculture, Feature Importance, Cross-Validation

1. Introduction

1.1 Purpose of the Project

Soil is the foundation of agricultural productivity, and its quality directly influences crop yield, resource utilization and long-term land sustainability. Traditionally, soil quality assessment relies on manual laboratory testing and expert interpretation, which is time-consuming, resource-intensive and often inaccessible to small and marginal farmers. Because soil health depends on several interacting parameters such as pH, nitrogen, phosphorus, potassium, organic matter and moisture, manual evaluation struggles to scale and often delays timely agricultural decisions.

This project is motivated by the need for a fast, scalable and explainable alternative. The main stakeholders are farmers, agronomists, agricultural extension workers and land-management planners, who can use automated soil quality predictions for land and crop planning, smarter fertilizer use and better resource management. A machine learning approach learns patterns from historical soil samples and delivers near-instant predictions once soil parameters are entered, without added expert cost per sample.

1.2 Problem Statement

The problem is formulated as a supervised multi-class classification task. Given a soil sample described by pH, nitrogen (N), phosphorus (P), potassium (K), organic matter and moisture, the system predicts the soil quality class of the sample (for example Poor / Moderate / Good, as defined in the dataset used). The expected output is the predicted soil quality class for each sample, together with performance metrics that compare the Decision Tree and Random Forest models.

1.3 Existing Methods and Disadvantages

Conventional soil quality assessment depends on laboratory analysis and expert judgement. These methods are slow, costly and difficult to scale to large numbers of samples, and they delay decisions that farmers need to make

in time. Fixed rule-based or threshold-based interpretation can also miss interactions among soil parameters, since soil health is determined by several attributes acting together.

Among data-driven methods, a single Decision Tree is simple and interpretable but tends to overfit, is sensitive to small changes in the data and has high variance. Ensemble methods such as Random Forest reduce this variance but are less transparent unless feature-importance analysis is added. All such models still require careful preprocessing, missing-value handling, leakage prevention and proper validation before they can be trusted for real-world agricultural use.

1.4 Proposed System

The proposed system follows the pipeline: soil dataset → data cleaning and missing-value imputation → encoding and feature engineering → train/test split → Decision Tree (Gini and entropy criteria, with pruning) → Random Forest (bagging and feature subsampling) → cross-validation → evaluation using accuracy, precision, recall, F1-score and confusion matrix → feature-importance analysis. The same data split and cross-validation scheme are used for both models so that the comparison is fair. The expected contribution is a reproducible, interpretable soil quality classification workflow and an evidence-based comparison between a single tree and an ensemble.

1.5 Research Objectives(atleast 4)

1. Collect and prepare a soil dataset containing pH, N, P, K, organic matter and moisture with a soil quality label.
2. Perform preprocessing (cleaning, missing-value imputation, encoding) and exploratory data analysis.
3. Engineer and select relevant features while preventing data leakage.
4. Build a Decision Tree classifier using Gini impurity and entropy criteria, with pruning to control overfitting.
5. Train a Random Forest ensemble using bagging and feature subsampling, and compare it with the Decision Tree using accuracy, precision, recall, F1-score, confusion matrix and cross-validation.
6. Analyse feature importance to identify the most influential soil attributes and discuss the practical applicability of the best model.

2. Literature Survey (recent studies from 2-3 years)

Review ten recent and directly relevant studies. Verify bibliographic details, DOI, volume, issue, pages/article number, and year before final submission. NOTE: the rows below are topic-level drafts; replace the Title and Reference columns with the real papers you select and adjust the other columns to match each paper.

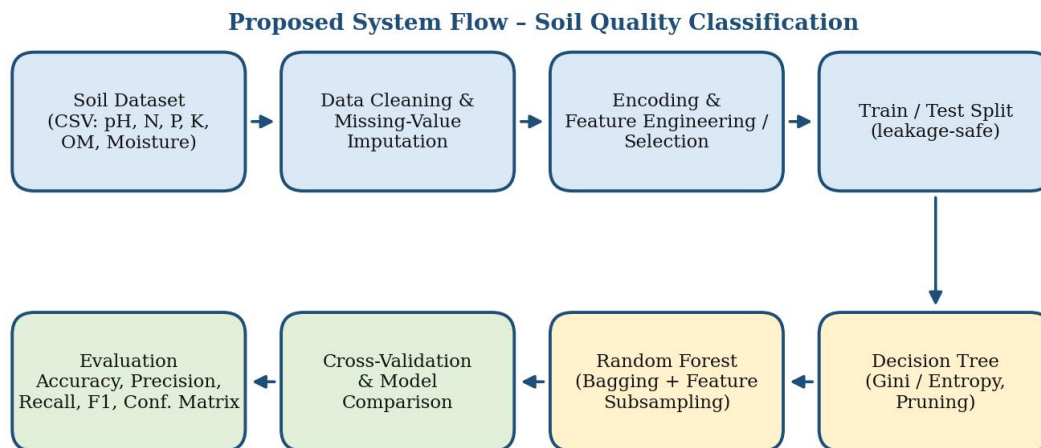
S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	Soil fertility classification using machine learning	[Add authors, year, journal/conference, volume/pages, DOI]	Manual lab testing is slow and not scalable	Classify soil fertility from nutrient and pH data	Supervised classifiers on N, P, K, pH, organic carbon	ML can automate fertility grading from soil test values	Validate on field data from more regions
2	Decision Tree models for soil quality assessment	[Add authors, year, journal/conference, volume/pages, DOI]	Need interpretable rules for farmers	Build transparent soil quality classifiers	Decision Tree; Gini / entropy splitting	Trees give readable rules but can overfit	Apply pruning and cross-validation
3	Random Forest for soil property classification	[Add authors, year, journal/conference, volume/pages, DOI]	Single models show high variance	Improve robustness of soil classification	Random Forest; bagging; feature subsampling	Ensembles generalize better than single trees	Study explainability of ensembles
4	Soil nutrient (N, P, K) based quality prediction	[Add authors, year, journal/conference, volume/pages, DOI]	Nutrients interact and are hard to judge manually	Predict quality from macronutrient levels	Classification on N, P, K and organic matter	Nutrient variables are strong predictors	Add micronutrients and weather data
5	Bagging and ensemble	[Add authors, year, journal/conference, volume/pages, DOI]	Unclear gain of ensembles over	Compare ensemble and	Bagging; Random	Ensembles are more accurate	Test boosting methods as well

	methods versus single trees	volume/pages, DOI]	one tree	single-tree performance	Forest; Decision Tree	and stable	
6	Pruning and overfitting control in decision trees	[Add authors, year, journal/conference, volume/pages, DOI]	Unpruned trees memorize training data	Control complexity and bias–variance trade-off	Pre-pruning; cost-complexity pruning	Pruning improves test-set generalization	Tune depth and leaf size systematically
7	Missing-value imputation for soil datasets	[Add authors, year, journal/conference, volume/pages, DOI]	Soil records often have missing readings	Handle missing values without bias	Mean/median and model-based imputation	Imputation choice affects model accuracy	Compare advanced imputation methods
8	Feature importance for soil quality indicators	[Add authors, year, journal/conference, volume/pages, DOI]	Stakeholders need to know key soil factors	Rank influential soil attributes	Gini importance; permutation importance	Few attributes drive most of the prediction	Add SHAP-style local explanations
9	Cross-validation and data leakage in agricultural ML	[Add authors, year, journal/conference, volume/pages, DOI]	Leakage inflates reported accuracy	Obtain reliable performance estimates	Stratified k-fold CV; leakage-safe pipelines	Proper validation gives trustworthy results	Use external or spatial validation
10	ML decision support for precision agriculture	[Add authors, year, journal/conference, volume/pages, DOI]	Models rarely reach practical farm use	Move soil prediction toward deployment	Decision-support tools; web/mobile apps	Fast predictions can guide fertilizer and crop plans	Integrate IoT soil sensors and real-time scoring

3. Methodology

3.1 Proposed System Flow Diagram

Proposed system flow for the soil quality classification project:



Final output: best-performing model and predicted soil quality class

Soil Dataset → Data Cleaning & Missing-Value Imputation → Encoding & Feature Engineering/Selection → Train/Test Split → Decision Tree (Gini/Entropy, Pruning) → Random Forest (Bagging + Feature Subsampling) → Cross-Validation → Evaluation → Best Model & Feature Importance.

3.2 Dataset Description

Dataset: [dataset name/source and file name]. The dataset is a CSV file of soil samples containing the parameters pH, nitrogen (N), phosphorus (P), potassium (K), organic matter and moisture, with soil quality as the target class. [Add the number of records, final feature count, class distribution, feature types and collection details taken from the executed dataset file.]

3.3 Experimental Setup

Python 3.10 or above on Windows 10/11, macOS or Linux, using Jupyter Notebook / Google Colab / VS Code. NumPy and Pandas are used for data handling and preprocessing, scikit-learn for the Decision Tree, Random Forest, cross-validation and evaluation metrics, and Matplotlib and Seaborn for EDA and result visualization. [Record exact hardware, Python version and library versions from the environment used for the final run.]

3.4 Data Preprocessing

- Data inspection and schema validation (soil parameter columns and target label)
- Duplicate removal
- Missing-value analysis and imputation for soil parameters
- Outlier analysis/treatment where justified
- Categorical encoding (e.g., soil quality labels or any categorical soil attributes)
- Numerical scaling/standardization where required (not mandatory for tree-based models)
- Target-label preparation
- Stratified train/test split and cross-validation, with all preprocessing fitted on training data only to prevent data leakage

3.5 Exploratory Data Analysis

EDA covers the distribution of pH, N, P, K, organic matter and moisture; soil quality class balance; correlation analysis among soil parameters; box plots and scatter plots of each parameter against soil quality class; and outlier inspection. [Insert the actual plots generated from the dataset.]

3.6 Feature Engineering and Feature Selection (if applicable)

Feature engineering includes imputing missing values, encoding categorical fields and, where justified, deriving features such as nutrient ratios (e.g., N:P:K). Feature selection is guided by correlation analysis and feature-importance scores from the trees, while the target label is never used to construct predictive features. [List selected and rejected features and the reasons.]

3.7 Model Building and Training

Baseline model: a Decision Tree classifier built with Gini impurity and entropy splitting criteria, with pruning (e.g., max_depth, min_samples_split, min_samples_leaf, cost-complexity pruning) to control overfitting.

Proposed model: a Random Forest ensemble trained with bagging (bootstrap sampling) and random feature subsampling at each split, with n_estimators and random_state fixed. Both models use impurity reduction as the splitting objective and are trained on the same training set and validated using cross-validation. [Record final parameter values.]

3.8 Hyperparameter Tuning(if applicable)

Hyperparameter tuning uses grid or random search with stratified k-fold cross-validation on the training data only.

Search space: for the Decision Tree — criterion, max_depth, min_samples_split, min_samples_leaf, ccp_alpha; for Random Forest — n_estimators, max_depth, max_features, min_samples_leaf. [Record the search space actually used and the selected parameters.]

3.9 Model Evaluation

Classification metrics: Accuracy, Precision, Recall, F1-score (macro/weighted if there are more than two classes) and the confusion matrix for each model on the held-out test set. Cross-validation mean $\pm$ standard deviation is reported to show stability. Feature-importance scores are used to interpret which soil attributes drive predictions. ROC-AUC may be added if applicable.

4. Results and Discussion

4.1 Preprocessing Results

[Report missing values before/after imputation, dataset dimensions, encoding results, final feature count and train/test sizes, copied from the actual run.]

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-score	CV Score (mean $\pm$ SD)
Existing study	[Value]	[Value]	[Value]	[Value]	[Value]
Decision Tree (Gini)	[Value]	[Value]	[Value]	[Value]	[Value]
Decision Tree (Entropy / pruned)	[Value]	[Value]	[Value]	[Value]	[Value]
Random Forest (default)	[Value]	[Value]	[Value]	[Value]	[Value]
Random Forest (tuned)	[Value]	[Value]	[Value]	[Value]	[Value]

4.3 Visual Results and discussion

[Insert figures from the actual run: class distribution and EDA plots, confusion matrices for the Decision Tree and Random Forest, the pruned Decision Tree visualization, cross-validation score comparison and the feature-importance plot.]

[Interpret the results: compare the Decision Tree and Random Forest using the measured metrics, explain why the better model performed well (e.g., variance reduction through bagging), discuss misclassified classes from the confusion matrices, the bias–variance trade-off, generalization to unseen soil samples, and which soil attributes are most influential.]

5. Conclusion

This project presents a machine learning approach for soil quality classification using Decision Tree and Random Forest. Soil parameters (pH, N, P, K, organic matter, moisture) were preprocessed, imputed, encoded and analysed before model training, and both models were evaluated using accuracy, precision, recall, F1-score, confusion matrix and cross-validation. [State the best model and its measured performance.] Feature-importance analysis highlights the most influential soil attributes, giving interpretable insight to agricultural stakeholders and supporting land and crop planning and fertilizer management.

6. Future Work

- Use larger and more diverse soil datasets from different regions and seasons.
- Evaluate additional algorithms such as Gradient Boosting, XGBoost and SVM.
- Perform deeper hyperparameter optimization.
- Investigate explainable AI techniques such as SHAP for sample-level explanations.
- Add micronutrients, weather and remote-sensing data; evaluate robustness and drift.
- Develop a deployable web/mobile application or IoT-based real-time soil quality predictor for farmers.
- Validate the approach on external field datasets.

7. References

References should contain the verified papers and technical sources actually used in the final report. The supplied project PPT contains no bibliographic records or DOI information, so this report does not invent author names or DOI numbers. Add the verified recent literature sources selected for the literature-survey table before final academic submission.

Signature of the Guide

Course Instructor